



Optimizing Logistics Through Data Intelligence

A Unified Data Architecture for Evergreen Marine Corp



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Introduction:

Logistics is the backbone of global trade, involving the efficient planning, movement, and storage of goods from origin to destination. For companies like **Evergreen Marine Corp**, a global logistics and shipping giant, managing this complex network means navigating vast data from multiple touchpoints—shipping ports, warehouses, delivery fleets, customs, and currency exchanges.

In today's highly competitive and time-sensitive logistics landscape, relying on siloed or delayed information can lead to shipment delays, increased costs, poor route planning, and missed revenue opportunities. To overcome these challenges, our project aims to build a robust, data-driven solution that integrates and analyzes operational, transactional, IoT, and external data in real time.

This system enables Evergreen to:

- Optimize delivery routes for faster, fuel-efficient operations,
- Track financial performance with up-to-date currency and transaction data,
- Monitor driver and fleet KPIs for performance benchmarking, and
- Provide live, interactive dashboards to support strategic decision-making.

By transforming scattered data into centralized intelligence, this project empowers Evergreen Marine Corp to achieve greater transparency, agility, and sustainability in its logistics operations—meeting both current demands and future growth.

Mission and Vision:

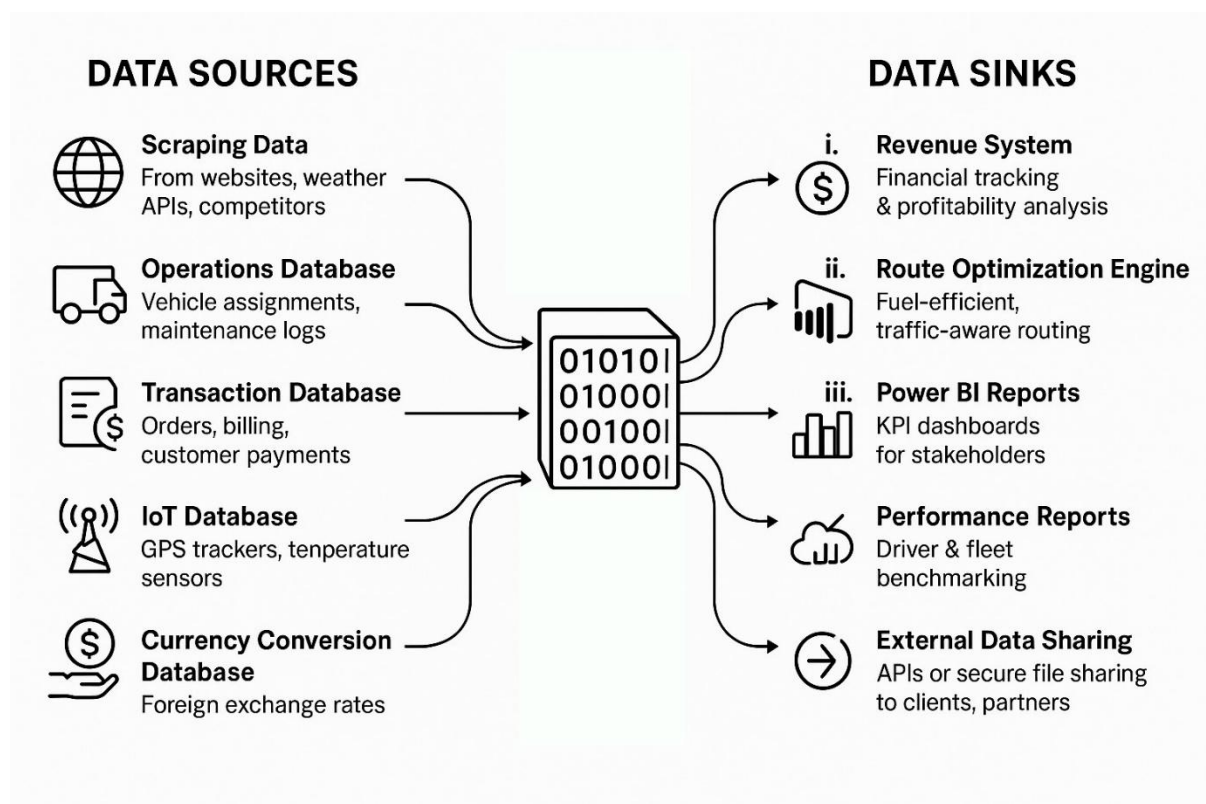
Mission

To collect, integrate, and analyze diverse operational, transactional, and external data in real-time, enabling efficient route optimization, financial tracking, and performance benchmarking for smarter logistics management.

Vision

To revolutionize logistics through intelligent, real-time data integration—delivering seamless operations, optimized performance, and unmatched transparency for every stakeholder.

Vision Diagram



Objectives

Enable real-time, data-driven decision-making by integrating diverse sources for optimized routing, financial accuracy, and performance insights.

Enhance transparency and stakeholder value through automated reporting, KPI dashboards, and secure external data sharing.

1. Optimize Routing by 20% within 6 Months:

Leverage the route optimization engine to reduce fuel usage and improve delivery time by at least 20% by the end of Q2.

2. Achieve 95% Accuracy in Financial Tracking by Q4:

Integrate currency conversion and transactional data to ensure the revenue system maintains <5% deviation in profitability analysis.

3. Deploy Real-Time KPI Dashboards by Next Quarter:

Implement Power BI dashboards using live IoT and operations data to monitor driver, fleet, and delivery KPIs accessible to all key stakeholders.

4. Benchmark Driver & Fleet Performance Monthly:

Generate automated performance reports with metrics (e.g., speed, fuel use, delivery punctuality) and distribute to fleet managers on a monthly basis.

5. Enable Secure External Data Sharing by End of Year:

Establish secure API endpoints or encrypted data sharing protocols to allow clients and partners to access necessary logistics performance data.

DATA SOURCES

Scraping Data

- **Nature:** Unstructured or semi-structured data scraped from carrier sites, weather updates, traffic APIs, geopolitical think tanks, customer interactions, shipment tracking requests, and form submissions.
- **Format:** JSON, HTML, CSV, web stream
- **Usage:** Used to make routing decisions, delivery estimates, competitor benchmarking, enhance user experience, demand forecasting, and lead generation.

In-House Data

- **Nature:** Structured data maintained internally, including vehicle assignments, maintenance logs, route schedules, staff rosters, and delivery calendars.
- **Format:** Relational tables (SQL)
- **Usage:** Supports fleet optimization, internal resource planning, maintenance scheduling, and fuel efficiency tracking.

Website Transaction Data

- **Nature:** Structured records of customer interactions through the company's website, including shipment orders, digital payments, service bookings, and billing information.
- **Format:** SQL tables or NoSQL documents
- **Usage:** Used for invoicing, real-time financial tracking, customer engagement analytics, and e-commerce insights.

IoT Database

- **Nature:** Time-series telemetry data from GPS trackers, onboard sensors, and connected fleet devices.
- **Format:** JSON, CSV, or time-series database formats
- **Usage:** Enables real-time tracking, environmental monitoring (e.g., temperature-sensitive cargo), and performance analytics.

Currency Conversion Database

- **Nature:** Structured foreign exchange rate data used for international logistics operations.
- **Format:** JSON, XML, or SQL
- **Usage:** Powers multi-currency invoicing, cost estimation, pricing strategy, and global budgeting.

DATA SINKS

Revenue System

- **Purpose:** To calculate, track, and analyze all financial transactions across Evergreen's operations.
- **Details:** Ingests processed data from website transactions, currency conversions, and customer billing to provide real-time visibility into revenue streams, cost margins, and profitability by route, region, or service type. Integrated with accounting tools for auditing and forecasting.

Route Intelligence

- **Purpose:** To improve delivery efficiency and reduce operational costs by determining the most optimal routes.
- **Details:** Uses enriched data from IoT sensors (e.g., GPS, traffic, load status), in-house fleet operations, and external data sources (e.g., weather, road conditions). Applies algorithms or machine learning models to optimize routes dynamically for time, fuel efficiency, and vehicle utilization.

Dashboard Reports

- **Purpose:** To provide interactive dashboards and business intelligence for cross-departmental decision-making.
- **Details:** Consolidates cleaned and structured data from all sources—revenue, fleet, IoT, website transactions—into analytical models and Power BI dashboards. Visualizes KPIs like delivery success rate, shipment delays, operational costs, and financial performance.

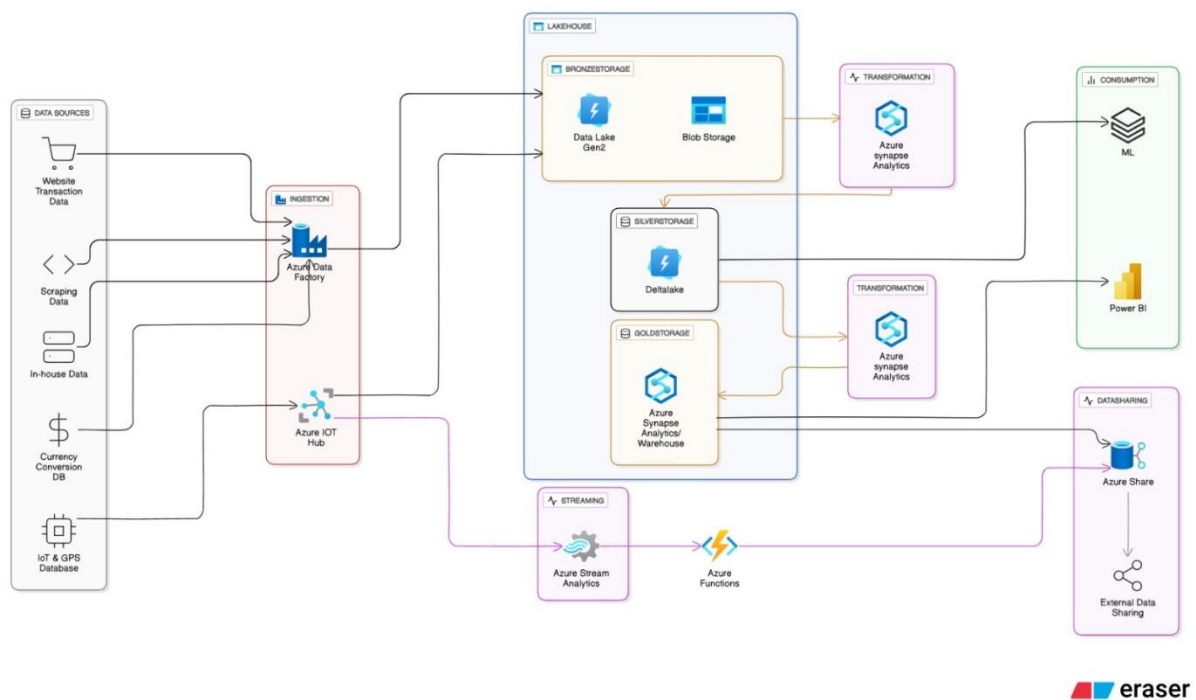
Ship Performance Report

- **Purpose:** To monitor, evaluate, and benchmark logistics and fleet efficiency over time.
- **Details:** Generated on daily, weekly, or monthly intervals using IoT metrics (idle time, fuel usage), delivery logs, and SLA compliance data. Enables operations teams to identify trends, flag inefficiencies, and implement improvements.

External Data Sharing

- **Purpose:** To securely distribute selected data with clients, partners, regulatory bodies, or third-party services.
- **Details:** Shares curated datasets (e.g., delivery status, emissions data, service KPIs) via APIs or secure file transfers. Information is anonymized or filtered to maintain confidentiality. Supports transparency, compliance, and collaborative analytics.

Cloud Architecture:



This cloud architecture diagram represents a modern data platform using various Azure services to manage data ingestion, storage, transformation, streaming, sharing, and consumption. It follows the Lakehouse architecture, which combines elements of data lakes and data warehouses for a unified data processing approach.

Architecture Breakdown

1. Data Sources

The architecture ingests data from diverse sources:

- Website Transaction Data
- Scraping Data
- In-house Data
- Currency Conversion Database
- IoT & GPS Database

These sources represent a mix of **structured, semi-structured, and unstructured** data originating from transactional systems, external APIs, and sensor networks.

2. Ingestion Layer

Handled using:

- **Azure Data Factory (ADF)** – ETL/ELT tool for ingesting data from batch and streaming sources into the data platform.
- **Azure IoT Hub** – Specifically designed to ingest real-time telemetry data from IoT devices.

These services ensure seamless ingestion into the data storage layers for both real-time and batch processing.

3. Lakehouse Storage

Organized into **Bronze, Silver,** and **Gold** layers:

- **Bronze Storage:**
 - Raw data stored in **Azure Data Lake Gen2** and **Blob Storage**.
 - Acts as the first landing zone for all incoming data.
- **Silver Storage:**
 - Data is cleaned, structured, and stored in **Delta Lake** (enabling ACID transactions on large-scale data).

- **Gold Storage:**
 - Curated, business-ready data stored in **Azure Synapse Analytics (Warehouse)**.
 - Supports analytics, reporting, and downstream processing.

4. Transformation

- **Azure Synapse Analytics** is used throughout the architecture for:
 - Querying
 - Transforming data (from Bronze to Silver to Gold)
 - Running complex analytics workflows

This central analytics engine connects tightly with the Lakehouse and supports both SQL and Spark-based processing.

5. Streaming

- For **real-time data pipelines**, the architecture includes:
 - **Azure Stream Analytics** – Real-time stream processing.
 - **Azure Functions** – Event-driven compute service to trigger downstream processes or custom logic.

Ideal for time-sensitive insights from IoT data or streaming APIs.

6. Consumption Layer

- **Machine Learning (ML)** models and **Power BI** dashboards use the curated data for:
 - Predictive analytics
 - Reporting and visualization
 - Business intelligence applications

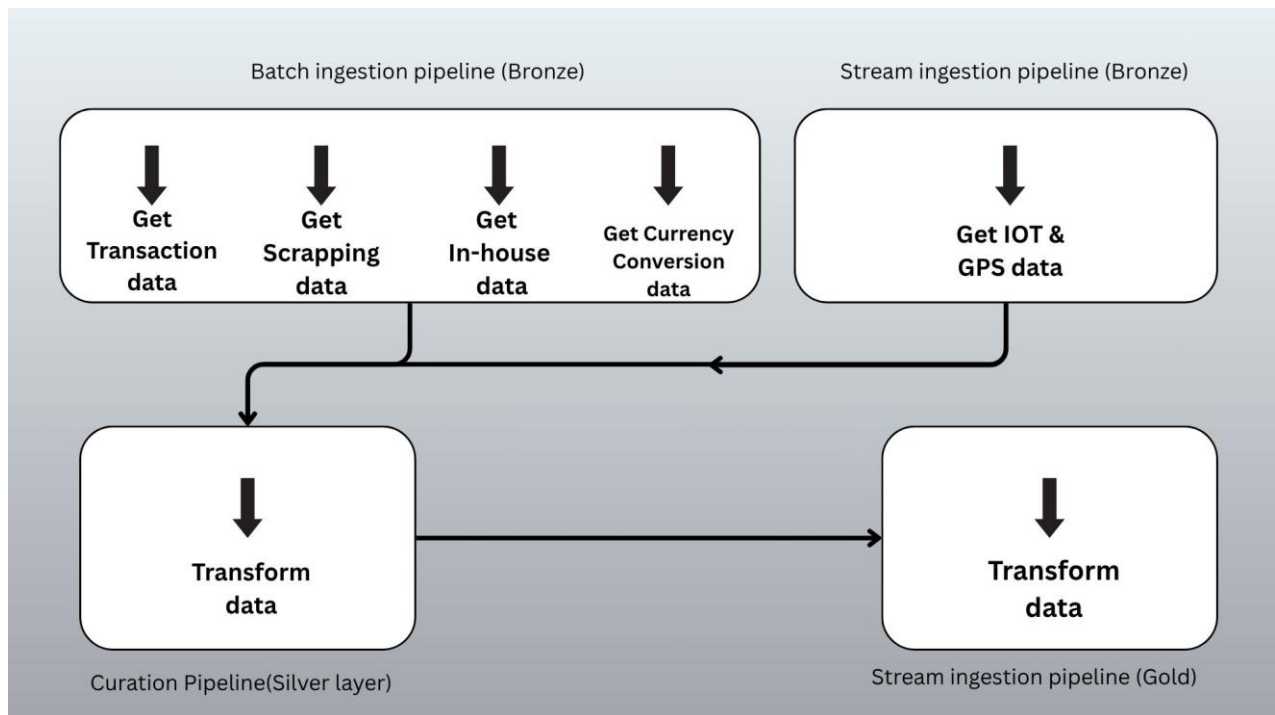
7. Data Sharing

- **Azure Share** facilitates secure sharing of data with external consumers.
- Integrates with external systems for **governed, policy-driven data sharing**.

Key Benefits

- **Modular Design:** Clear separation of concerns between ingestion, storage, transformation, and consumption.
- **Scalability:** Leverages cloud-native services to scale on-demand.
- **Flexibility:** Supports both batch and streaming data.
- **Governance:** Uses Delta Lake and Synapse for data integrity and auditing.
- **Integration:** Unified data platform that connects with ML, BI, and external systems seamlessly.

Pipeline Design



While the architecture sets the stage, the **data pipeline** is where the real transformation happens.

This diagram illustrates a simplified view of the data ingestion and transformation workflow across Bronze, Silver, and Gold layers in a data platform, splitting the flow into batch and stream pipelines.

Batch Ingestion (Bronze to Silver)

- **Sources:** Transactions, scraping, in-house data, currency conversion databases.
- **Flow:** Raw data lands in the Bronze layer → passes through the transformation stage → emerges in the Silver layer as clean, standardized datasets.

Stream Ingestion (Bronze to Gold)

- **Sources:** IoT & GPS data.
- **Flow:** Live data streams into Bronze → instantly transformed → pushed to the Gold layer for immediate reporting and decision-making.

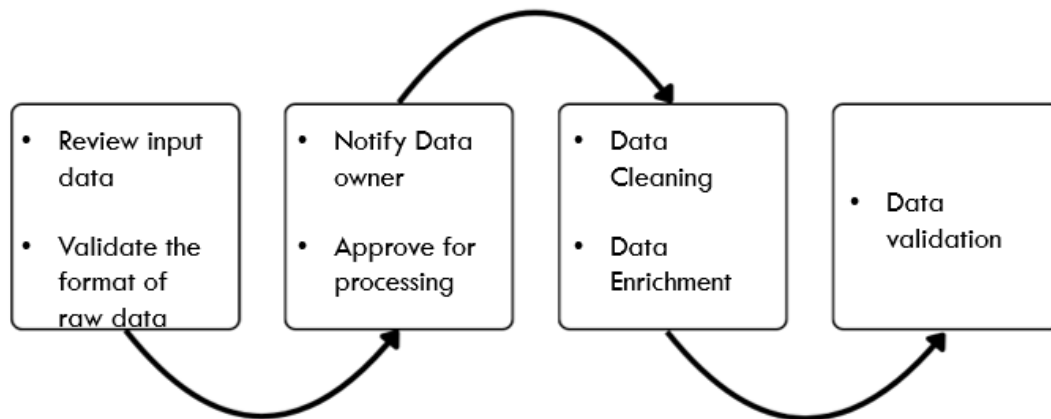
Transformation Stages

- **Silver Layer:** Removes errors, standardizes formats, and integrates datasets for consistency.
- **Gold Layer:** Produces business-ready aggregates, KPIs, and models optimized for BI and predictive analytics.

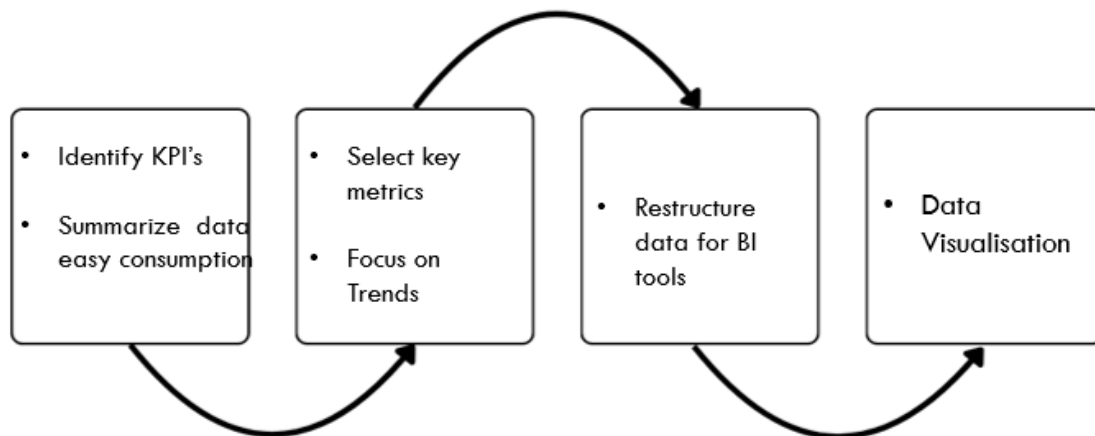
By combining the Azure Lakehouse architecture with the Bronze–Silver–Gold data pipeline pattern, businesses can move from chaotic raw data to trusted, actionable intelligence — all within a single, scalable cloud platform.

In a world where the speed of insight can be the difference between leading and lagging, this modern approach ensures organizations are not just collecting data, but truly using it to shape their future.

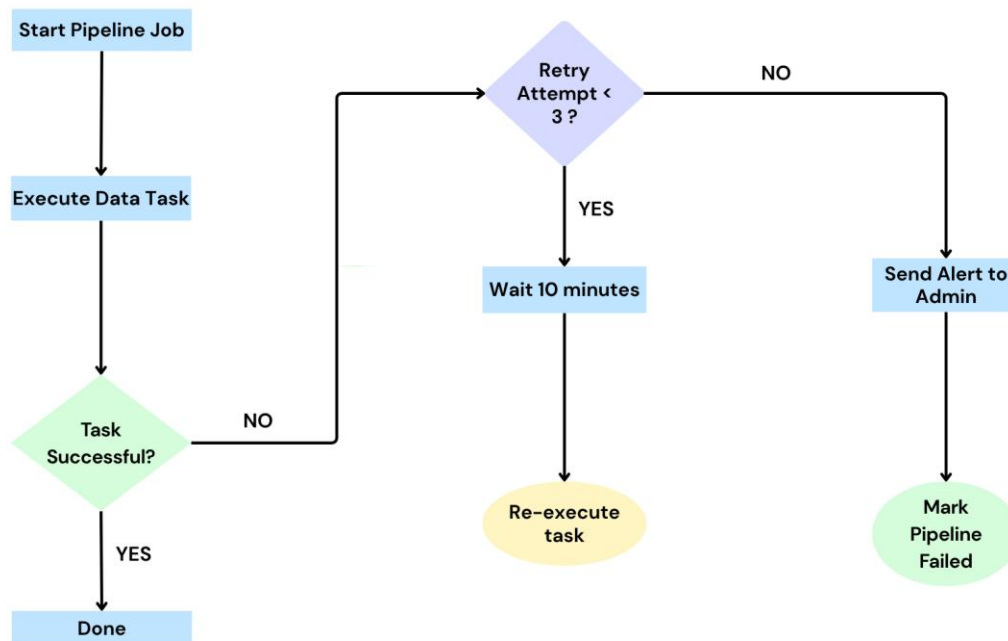
Data Curation (Silver Layer):



Data Aggregation (Gold layer):



Pipeline Failure Strategy:



This diagram represents a **pipeline job execution and retry workflow** — essentially a safeguard system to ensure that data tasks either complete successfully or fail gracefully with alerts.

Step-by-Step Flow Explanation

1. Start Pipeline Job

- The process begins when the pipeline job is triggered (either manually or on a schedule).

2. Execute Data Task

- The main data processing task is run — this could be ingestion, transformation, or loading data into storage.

3. Check Task Success

- If the task runs successfully, the process moves straight to **Done**.

4. If Task Fails → Retry Logic

- The system checks whether the number of retry attempts is less than 3.

- YES → Wait for 10 minutes (cool-off period to handle transient issues), then Re-execute Task.
- NO → Skip retry and go straight to failure handling.

5. Failure Handling

- If all retries are exhausted:
 - Send Alert to Admin so the issue can be investigated.
 - Mark Pipeline as Failed in the system for logging and monitoring.

Purpose in the Larger Data Architecture

In the Azure Lakehouse + Bronze–Silver–Gold pipeline context we discussed earlier, this diagram represents the automation and fault-tolerance built into your ingestion or transformation pipelines.

It ensures:

- Reliability — transient issues get retried automatically.
- Operational Efficiency — admins are alerted only when necessary.
- Data Integrity — failed runs are logged, preventing partial or corrupted data from moving forward.